The documentation and process conversion measures necessary to comply with this revision shall be completed by 30 March 1994

INCH-POUND

MIL-S-19500/396D
30 December 1993
SUPERSEDING
MIL-S-19500/396C
6 February 1984

MILITARY SPECIFICATION
SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, SWITCHING
TYPES 2N3762, 2N3762L, 2N3763, 2N3763L, 2N3764, AND 2N3765
JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for PNP silicon switching transistors. Three levels of product assurance is provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See 3.3 (similar to TO-5, TO-39, and TO-46).

1.3 Maximum ratings.

| Types | $P_T$ $T_A$ = +25°C | $V_{CBO}$ | $V_{CEO}$ | $V_{EBO}$ | $I_C$ | $T_{OP}$ and $T_{STG}$ | $R_{\theta JC}$ |
|---|---|---|---|---|---|---|---|
| | W | V dc | V dc | V dc | A dc | °C | °C/W |
| 2N3762 | 1.0 1/ | 40 | 40 | 5 | 1.5 | -55 to +200 | 60 |
| 2N3762L | 1.0 1/ | 40 | 40 | 5 | 1.5 | -55 to +200 | 60 |
| 2N3763 | 1.0 1/ | 60 | 60 | 5 | 1.5 | -55 to +200 | 60 |
| 2N3763L | 1.0 1/ | 60 | 60 | 5 | 1.5 | -55 to +200 | 60 |
| 2N3764 | 0.5 2/ | 40 | 40 | 5 | 1.5 | -55 to +200 | 88 |
| 2N3765 | 0.5 2/ | 60 | 60 | 5 | 1.5 | -55 to +200 | 88 |

1/ Derate linearly at 5.71 mW/°C above $T_A$ = +25°C.
2/ Derate linearly at 2.86 mW/°C above $T_A$ = +25°C.

1.4 Primary electrical characteristics $T_A$ = +25°C. (Applies to all devices unless otherwise indicated.)

| Limits | $h_{FE1}$ $V_{CE}$ = 1.0 V dc $I_C$ = 10 mA dc | $h_{FE3}$ $V_{CE}$ = 1.0 V dc $I_C$ = 500 mA dc | $h_{FE5}$ 1/ $V_{CE}$ = 5.0 V dc $I_C$ = 1.5 A dc | |
|---|---|---|---|---|
| | | | 2N3762 2N3763 2N3762L 2N3763L 2N3764 | 2N3765 |
| Min | 35 | 40 | 30 | 20 |
| Max | | 140 | | |

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: NASA/Parts Project Office (NPPO), NASA Goddard Space Flight Center, Code 310.A, Greenbelt, MD 20771 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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| Limits | $\lvert h_{FE} \rvert$ f = 100 MHz $V_{CE}$ = 10 V dc $I_C$ = 50 mA dc | | $V_{CE(SAT)}$3 $I_C$ = 500 mA dc $I_B$ = 50 mA dc 1/ | $C_{obo}$ $V_{CE}$ = 10 V dc $I_E$ = 0 100 kHz ≤ f ≤ 1 MHz | Pulse response | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | See fig.1 | | See fig. 2 | |
| | 2N3762 2N3764 | 2N3763 2N3765 | | | $t_d$ | $t_r$ | $t_s$ | $t_f$ |
| | | | V dc | pF | ns | ns | ns | ns |
| Min | 1.8 | 1.5 | | | | | | |
| Max | 6.0 | 6.0 | 0.5 | 15 | 8 | 35 | 80 | 35 |

1/ Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, appendix F, figure 8, T-1C (for 2N3762 and 2N3763, similar to TO-39) and T-1A (for 2N3762L and 2N3763L, similar to TO-5) and figure 9, T-2B (for 2N3764 and 2N3765, similar to TO-46).

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-S-19500.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.



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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3 Screening (JANTX, JANTXV, and JANS levels). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table II of MIL-S-19500) | Measurements | |
|---|---|---|
| | JANS level | JANTX and JANTXV levels |
| 9 | $I_{CBO}$ and $h_{FE3}$ | Not applicable |
| 11 | $I_{CBO}$; $h_{FE3}$, $\Delta ICBO$ = 100% of initial value or 10 nA dc whichever is greater; $\Delta h_{FE3}$ = ±15% | $I_{CBO}$ and $h_{FE3}$ |
| 12 | See 4.3.1 | See 4.3.1 |
| 13 | Subgroups 2 and 3 of table I herein; $\Delta I_{CBO}$ = 100% of initial value or 10 nA dc, whichever is greater; $\Delta h_{FE3}$ = ±15% | Subgroup 2 of table I herein; $\Delta I_{CBO}$ = 100% of initial value or 10 nA dc, whichever is greater; $\Delta h_{FE3}$ = ±15% |

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

$T_A$ = Room ambient as defined in the general requirements of MIL-STD-750, 4.5 (all product assurance levels);

JANS level:

2N3762 - $V_{CB}$ = 10 V dc $P_T$ = 1.0 W
2N3763 - $V_{CB}$ = 10 V dc $P_T$ = 1.0 W
2N3764 - $V_{CB}$ = 10 V dc $P_T$ = 0.5 W
2N3765 - $V_{CB}$ = 10 V dc $P_T$ = 0.5 W

JANTX and JANTXV levels:

2N3762 - $V_{CB}$ = 30 V dc $P_T$ = 1.0 W
2N3763 - $V_{CB}$ = 40 V dc $P_T$ = 1.0 W
2N3764 - $V_{CB}$ = 30 V dc $P_T$ = 0.5 W
2N3765 - $V_{CB}$ = 40 V dc $P_T$ = 0.5 W

NOTE: No heat sink or forced air cooling on the devices shall be permitted. Power burn-in conditions for "L" suffix devices are identical to their corresponding non-L suffix devices.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.



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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500, and as follows. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps and footnotes of table II herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 2037 | Test condition A. |
| B4 | 1037 | $V_{CB}$= 10 V dc, $P_T$ = 1.0 W (for 2N3762, 2N3762L, 2N3763, and 2N3763L), $P_T$ = .5 W (for 2N3764 and 2N3765) at $T_A$ = +30°C ±5°C; $t_{on}$= $t_{off}$= 3 minutes minimum for 2,000 cycles. No heat sink or forced-air cooling on devices shall be permitted. |
| B5 | 1027 | $V_{CB}$= 10 V dc, $P_T$ = 1.0 W (for 2N3762, 2N3762L, 2N3763, and 2N3763L), $P_T$ = 0.5 W (for 2N3764 and 2N3765) at $T_A$ = +125°C ±25°C for 96 hours or $T_A$ = +100°C for 96 hours with $P_T$ adjusted as required by the chosen $T_A$ to give an average lot. $T_J$ = +275°C. Marking legibility requirements shall not apply. |
| B6 | 3131 | See 4.5.2. |

4.4.2.2 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 1027 | 2N3762, 2N3762L - $V_{CB}$= 30 V dc; $P_T$ = 1.0 W at $T_A$ = +30°C ±5°C<br>2N3763, 2N3763L - $V_{CB}$= 40 V dc; $P_T$ = 1.0 W at $T_A$ = +30°C ±5°C<br>2N3764 - $V_{CB}$= 30 V dc; $P_T$ = 0.5 W at $T_A$ = +30°C ±5°C<br>2N3765 - $V_{CB}$= 40 V dc; $P_T$ = 0.5 W at $T_A$ = +30°C ±5°C<br>No heat sink or forced-air cooling on the devices shall be permitted. |
| B5 | 3131 | See 4.5.2. |

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500 and as follows. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition A. |
| C6 | 1027 | 2N3762 - $V_{CB}$= 30 V dc; $P_T$ = 1.0 W at $T_A$ = +30°C ±5°C<br>2N3763 - $V_{CB}$= 40 V dc; $P_T$ = 1.0 W at $T_A$ = +30°C ±5°C<br>2N3764 - $V_{CB}$= 30 V dc; $P_T$ = 0.5 W at $T_A$ = +30°C ±5°C<br>2N3765 - $V_{CB}$= 40 V dc; $P_T$ = 0.5 W at $T_A$ = +30°C ±5°C<br>No heat sink or forced-air cooling on device shall be permitted. |

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.



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4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power application shall be 100 mA dc (2N3762, 2N3762L, 2N3763, and 2N3763L) and 60 mA dc (2N3764 and 2N3765).

b. Collector to base voltage magnitude shall be 10 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be +25°C $\leq T_R \leq$ +35°C and the chosen $T_R$ recorded before the test is started.

e. Mounting arrangement shall be with heat sink to case.

f. Maximum limit of $R_{\theta JC}$ shall be 60° C/W (2N3762, 2N3762L, 2N3763, and 2N3763L) and 88°C/W (2N3764 and 2N3765).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1).

b. Lead finish as specified (see 3.3.1).

c. Type designation and product assurance level.

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Breakdown voltage, collector to emitter | 3011 | Bias condition D; $I_C$ = 10 mA dc; Pulsed (see 4.5.1) | $V_{(BR)CEO}$ | | | |
| 2N3762, 2N3764 | | | | 40 | | V dc |
| 2N3763, 2N3765 | | | | 60 | | V dc |
| Breakdown voltage collector to base | 3001 | Bias condition D; $I_C$ = 10 µA dc | $V_{(BR)CBO}$ | | | |
| 2N3762, 2N3764 | | | | 40 | | V dc |
| 2N3763, 2N3765 | | | | 60 | | V dc |
| Breakdown voltage emitter to base | 3026 | Bias condition D; $I_E$ = 10 µA dc | $V_{(BR)EBO}$ | 5 | | V dc |
| Collector to base cutoff current | 3036 | Bias condition D; | $I_{CBO}$ | | 100 | nA dc |
| 2N3762, 2N3764 | | $V_{CB}$ = 20 V dc; | | | | |
| 2N3763, 2N3765 | | $V_{CB}$ = 30 V dc | | | | |
| Collector to emitter cutoff current | 3041 | Bias condition A; | $I_{CEX1}$ | | 100 | nA dc |
| 2N3762, 2N3764 | | $V_{EB}$ = 2.0 V dc; | | | | |
| 2N3763, 2N3765 | | $V_{CE}$ = 20 V dc; $V_{CE}$ = 30 V dc | | | | |
| Emitter to base cutoff current | 3061 | Bias condition D; $V_{EB}$ = 2.0 V dc | $I_{EBO}$ | | 200 | nA dc |
| Forward - current transfer ratio | 3076 | $V_{CE}$ = 1.0 V dc; $I_C$ = 10 mA dc | $h_{FE1}$ | 35 | | |
| Forward - current transfer ratio | 3076 | $V_{CE}$ = 1.0 V dc; $I_C$ = 150 mA dc; Pulsed (see 4.5.1) | $h_{FE2}$ | 40 | | |
| Forward - current transfer ratio | 3076 | $V_{CE}$ = 1.0 V dc; $I_C$ = 500 mA dc; Pulsed (see 4.5.1) | $h_{FE3}$ | 40 | 140 | |

See footnote at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 - Continued. | | | | | | |
| Forward - current transfer ratio | 3076 | $V_{CE}$ = 1.5 V dc; $I_C$ = 1.0 A dc; Pulsed (see 4.5.1) | $h_{FE4}$ | | | |
| 2N3762, 2N3764 | | | | 30 | 120 | |
| 2N3763, 2N3765 | | | | 20 | 80 | |
| Forward - current transfer ratio | 3076 | $V_{CE}$ = 5.0 V dc; $I_C$ = 1.5 A dc; Pulsed (see 4.5.1) | $h_{FE5}$ | | | |
| 2N3762, 2N3764 | | | | 30 | | |
| 2N3763, 2N3765 | | | | 20 | | |
| Collector to emitter voltage (saturated) | 3071 | $I_C$ = 10 mA dc; $I_B$ = 1 mA dc; Pulsed (see 4.5.1) | $V_{CE(SAT)1}$ | | 0.10 | V dc |
| Collector to emitter voltage (saturated) | 3071 | $I_C$ = 150 mA dc; $I_B$ = 15 mA dc; Pulsed (see 4.5.1) | $V_{CE(SAT)2}$ | | 0.22 | V dc |
| Collector to emitter voltage (saturated) | 3071 | $I_C$ = 500 mA dc; $I_B$ = 50 mA dc; Pulsed (see 4.5.1) | $V_{CE(SAT)3}$ | | 0.50 | V dc |
| Collector to emitter voltage (saturated) | 3071 | $I_C$ = 1.0 A dc; $I_B$ = 100 mA dc; Pulsed (see 4.5.1) | $V_{CE(SAT)4}$ | | 0.90 | V dc |
| Base to emitter voltage (saturated) | 3066 | Test condition A; $I_C$ = 10 mA dc; $I_B$ = 1 mA dc | $V_{BE(SAT)1}$ | | 0.80 | V dc |
| Base to emitter voltage (saturated) | 3066 | Test condition A; $I_C$ = 150 mA dc; $I_B$ = 15 mA dc; Pulsed (see 4.5.1) | $V_{BE(SAT)2}$ | | 1.0 | V dc |
| Base to emitter voltage (saturated) | 3066 | Test condition A; $I_C$ = 500 mA dc; $I_B$ = 50 mA dc; Pulsed (see 4.5.1) | $V_{BE(SAT)3}$ | | 1.2 | V dc |
| Base to emitter voltage (saturated) | 3066 | Test condition A; $I_C$ = 1.0 A dc; $I_B$ = 100 mA dc; Pulsed (see 4.5.1) | $V_{BE(SAT)4}$ | .90 | 1.40 | V dc |

See footnote at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 3 | | | | | | |
| High - temperature operation | | $T_A$ = +150°C | | | | |
| Collector to emitter cutoff current<br>2N3762, 2N3764<br>2N3763, 2N3765 | | Bias condition A;<br>$V_{EB}$ = 2 V dc;<br>$V_{CE}$ = 20 V dc;<br>$V_{CE}$ = 30 V dc | $I_{CEX2}$ | | 150 | µA dc |
| Low - temperature operation | | $T_A$ = -55°C | | | | |
| Forward - current transfer ratio | 3076 | $V_{CE}$ = 1.0 V dc;<br>$I_C$ = 500 mA dc;<br>Pulsed (see 4.5.1) | $h_{FE6}$ | 20 | | |
| Subgroup 4 | | | | | | |
| Magnitude of common emitter, small - signal short - circuit forward - current transfer ratio | 3306 | $V_{CE}$ = 10 V dc;<br>$I_C$ = 50 mA dc;<br>f = 100 MHz | $\|h_{fe}\|$ | | | |
| 2N3762, 2N3764 | | | | 1.8 | 6.0 | |
| 2N3763, 2N3765 | | | | 1.5 | 6.0 | |
| Open circuit output capacitance | 3236 | $V_{CB}$ = 10 V dc;<br>$I_E$ = 0;<br>100 kHz ≤ f ≤ MHz | $C_{obo}$ | | 15 | pF |
| Input capacitance (output open - circuited) | 3240 | $V_{EB}$ = .5 V dc;<br>$I_C$ = 0;<br>100 kHz ≤ f ≤ MHz | $C_{ibo}$ | | 80 | pF |
| Pulse response | | | | | | |
| Pulse delay time | 3251 | See figure 1 | $t_d$ | | 8 | ns |
| Pulse rise time | 3251 | See figure 1 | $t_r$ | | 35 | ns |
| Pulse storage time | 3251 | See figure 2 | $t_s$ | | 80 | ns |
| Pulse fall time | 3251 | See figure 2 | $t_f$ | | 35 | ns |

1/ For sampling plan, see MIL-S-19500. Electrical characteristics for "L" suffix devices are identical to their corresponding non-L suffix devices.



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TABLE II. Groups B and C electrical measurements. 1/ 2/ 3/

| Step | Inspection | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| 1. | Collector to base cutoff current<br>2N3762, 2N3764<br>2N3763, 2N3765 | 3036 | Bias condition D;<br>$V_{CB}$ = 20 V dc;<br>$V_{CB}$ = 30 V dc | $I_{CBO}$ | | <br>100<br>100 | <br>nA dc<br>nA dc |
| 2. | Collector to base cutoff current<br>2N3762, 2N3764<br>2N3763, 2N3765 | 3036 | Bias condition D;<br>$V_{CB}$ = 20 V dc;<br>$V_{CB}$ = 30 V dc | $I_{CBO}$ | | <br>200<br>200 | <br>nA dc<br>nA dc |
| 3. | Emitter to base cutoff current | 3061 | Bias condition D;<br>$V_{EB}$ = 2 V dc | $I_{EBO}$ | | 200 | nA dc |
| 4. | Collector to emitter voltage (saturated) | 3071 | $I_C$ = 500 mA dc;<br>$I_B$ = 50 mA dc;<br>pulsed (see 4.5.1) | $V_{CE(SAT)3}$ | | 0.5 | V dc |
| 5. | Base to emitter voltage (saturated) | 3066 | Test condition A;<br>$I_C$ = 500 mA dc;<br>$I_B$ = 50 mA dc;<br>pulsed (see 4.5.1) | $V_{BE(SAT)3}$ | | 1.2 | V dc |
| 6. | Forward - current transfer ratio | 3076 | $V_{CE}$ = 1.0 V dc;<br>$I_C$ = 500 mA dc;<br>pulsed (see 4.5.1) | $h_{FE3}$ | 40 | 140 | |
| 7. | Collector to base cutoff current<br>2N3762 & L, 2N3764<br>2N3763 & L, 2N3765 | 3036 | Bias condition D;<br>$V_{CB}$ = 20 V dc;<br>$V_{CB}$ = 30 V dc | $\Delta I_{CBO}$ | ±100% of initial value or 10 nA dc, whichever is greater | | |
| 8. | Forward - current transfer ratio | 3076 | $V_{CE}$ = 1.0 V dc;<br>$I_C$ = 500 mA dc | $\Delta h_{FE3}$ | ± 20% change from initial reading | | |
| 9. | Collector to emitter voltage (saturated) | 3071 | $I_C$ = 500 mA dc;<br>$I_B$ = 50 mA dc;<br>pulsed (see 4.5.1) | $\Delta V_{CE(SAT)3}$ | ±50 mV dc change from previous measured value | | |

1/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:

a. Subgroup 3, see table II herein, steps 1, 3, 4, 5, and 6.
b. Subgroups 4 and 5, see table II herein, steps 7, 8, and 9.

2/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

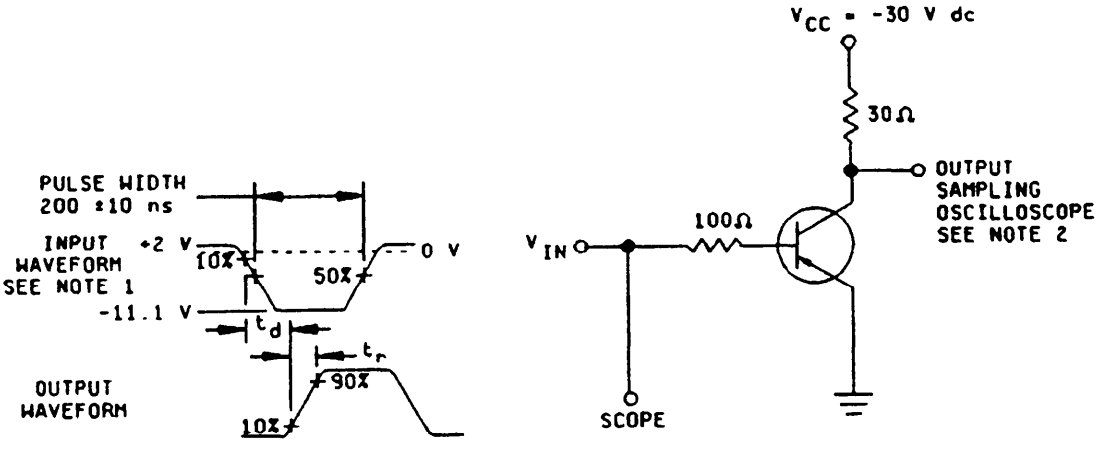
a. Subgroup 2, see table II herein, steps 1 and 5.
b. Subgroups 3 and 6, see table II herein, steps 2 and 8.

3/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroups 2 and 3, see table II herein, steps 1, 3, 4, 5, and 6 (JANS); and steps 1 and 6 (JANTX and JANTXV).
b. Subgroup 6, see table II herein, steps 2, 3, 4, 5, 6, 7, and 8 (JANS); and steps 2 and 8 (JANTX and JANTXV).

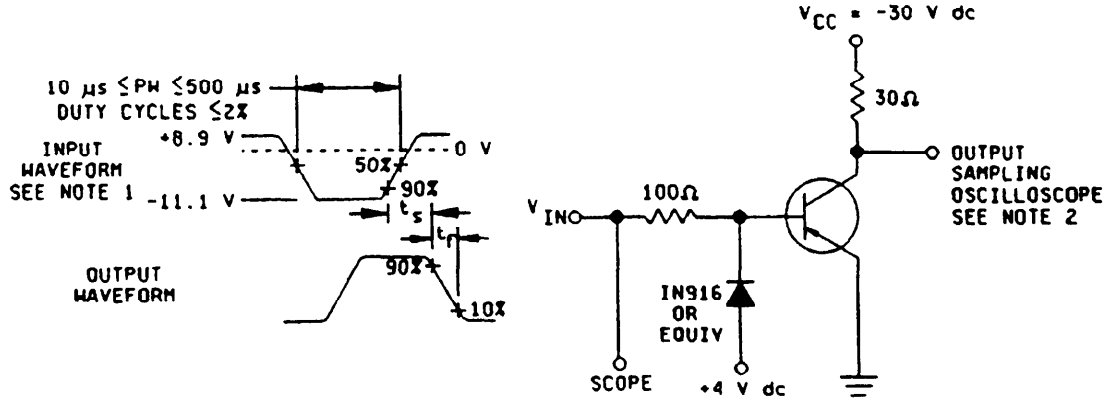


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NOTES:
1. The rise time ($t_r$) of the applied pulse shall be ≤ 2 ns, duty cycle ≤ 2%, and the generator source impedance shall be 50Ω.
2. Sampling oscilloscope: $Z_{in}$ ≥ 100 kΩ, $C_{in}$ ≤ 12 pF, rise time ≤ .1 ns.
3. $I_{B1}$ = -100 mA dc.

FIGURE 1. Pulse response test circuit for $t_d$ and $t_f$.




NOTES:
1. The rise time ($t_r$) of the applied pulse shall be ≤ 2 ns, duty cycle ≤ 2%, and the generator source impedance shall be 50Ω.
2. Sampling oscilloscope: $Z_{in}$ ≥ 100 kΩ, $C_{in}$ ≤ 12 pF, rise time ≤ .1 ns.
3. $I_{B1}$ = +$I_{B2}$ = -100 mA dc.

FIGURE 2. Pulse response test circuit for $t_s$ and $t_f$.



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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - MI
Navy - AS, CG, MC
Air Force - 15, 19, 85, 99
DLA - ES

Preparing activity:
NASA - NA

Agent:
DLA - ES

(Project 5961-1462)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

**INSTRUCTIONS**

The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-S-19500/396D | 2. DOCUMENT DATE (YYMMDD)<br>30 December 1993 |
|---|---|---|

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, SWITCHING TYPES 2N3762, 2N3762L, 2N3763, 2N3763L, 2N3764, AND 2N3765 JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>(If applicable) | 7. DATE SUBMITTED<br>(YYMMDD) |

8. PREPARING ACTIVITY

| a. NAME | b. TELEPHONE (Include Area Code)<br>(1) Commercial<br>301-731-8694 | (2) AUTOVON |
|---|---|---|
| c. ADDRESS (Include Zip Code)<br>NASA/Parts Project Office (NPPO)<br>NASA Goddard Space Flight Center<br>Code 310.A<br>Greenbelt, MD 20771 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |



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